

CSC 226 Exercise 3
An Nguyen, V01073373
31 Jan 2026

Problem:

Consider a hash table with size m (indices $0, 1, \dots, m - 1$) which uses chaining to resolve collisions. Suppose the following weird hash function is used:

$$h(k) = 1 + (k \bmod (m - 1))$$

Notice that the function $h(k)$ does not satisfy the uniform hashing assumption, since no element will ever hash to index zero. Suppose n randomly selected keys are inserted into the table. Derive a formula for the expected number of items in the list at index 1 of the table and justify your derivation.

Answer:

From the question, the hash function $h(k)$ above does not assign any keys to index 0, so it does not satisfy the Uniform Hashing Assumption (UHA). However, as mentioned in lecture, we can view the UHA as an assumption of the behaviour of an idealized hash function in our analysis, since the benefits of uniform hashing apply to some extent even in cases where a hash function is *almost* uniform (which is ours).

Claim:

Under the Uniform Hashing Assumption, the expected size of the list L_j at each index j of the table is $\frac{n}{m}$.

Justification:

Consider a randomly selected sequence $k_1, \dots, k_n$ of keys in U . Define a random variable X_{ij} as follows:

$$X_{ij} = \begin{cases} 1 & \text{if } h(k_i) = j \\ 0 & \text{otherwise} \end{cases}$$

By the UHA, we have:

$$P(X_{ij} = 1) = P(h(k_i) = j) = 1/m \text{ for } 1 \leq i \leq n \text{ and } 0 \leq j \leq m.$$

Since we are trying to find the expected number of items in the list at index 1, and considering the hash function:

$$h(k) = 1 + (k \bmod (m - 1)),$$

we can imply that a key hashes to index 1 if and only if $k \bmod (m - 1) = 0$. Values of $(k \bmod (m - 1))$ range from $0, 1, \dots, m - 2$, thus:

$$P(h(k_1) = 1) = \frac{1}{m - 1}$$

Therefore, the expected value of X_{i1} is:

$$E[X_{i1}] = 1 \cdot P(X_{i1} = 1) + 0 \cdot P(X_{i1} = 0) = \frac{1}{m-1}$$

The size of L_1 is then equal to the number of keys hashed to index 1, which is

$$K_1 = X_{11} + X_{21} + \dots + X_{n1}$$

so the expected size of L_1 is:

$$E[K_1] = E[X_{11} + X_{21} + \dots + X_{n1}] = E[X_{11}] + \dots + E[X_{n1}] = n \cdot \frac{1}{m-1}$$

Hence, the desired formula for expected size of the list L_j at index 1:

$$\boxed{E[K_1] = \frac{n}{m-1}}$$